Predictive Models in Sustainable Chemical Systems

In this lecture block we will study how predictive models support the development of sustainable chemical systems. Together we will:

- dive into the foundations of modelling
- develop data-driven models for predicting molecular properties
- apply the models for selecting solvents in line with green chemistry principles

Dr.-Ing. Laura König-Mattern

Group Leader for Computational Biorefining
Center for the Transformation of Chemistry
Mail: laura.koenig-mattern@ctc-germany.org
Homepage: **https://transforming-chemistry.org/forschung/forschungsgruppen/#biorefineries**

- background in **process engineering**
    - Bachelor in **Biosystems Engineering** (Otto von Guericke University Magdeburg)
    - Master in **Systems Engineering and Engineering Cybernetics** (Otto von Guericke University Magdeburg)
- PhD and research on **computer-aided molecular and process design for biorefineries**
- Guest PhD at EPFL in Lausanne

Course materials

All course materials including slides, data, and code can be found on GitHub: **Sustainable Chemical Systems Course**
Download the course contents. Navigate to "html_slides" and open them in your browser. If you prefer pdfs, check out the "pdf_export"-folder.

Installation

All installation instructions can be found in the README.md. Installation will be addressed later in the course as well.

Literature

Online Courses:

- Deep Learning Specialization by Andrew Ng, K. Katanforoosh and B. Mourri from Coursera learning platform: A course series that requires a minumum of mathematical skills but still brings you from zero to a very good understanding and practical training on deep learning.

Books:

- Deep Learning by I. Goodfellow, Y. Bengio and A. Courville: Open access, authored by pioneers on deep learning.
- Hands-On Machine Learning with Scikit-Learn and TensorFlow by A. Geron: For people who prefer learning by doing.

What is a model?

There are different definitions for "model" across disciplines
Philosophy
According to the philosopher Herbert Stachowiak, a model is characterized by at least three properties:

1. Mapping
   A model always is a model of something. It is an image or representation of some natural or artificial, existing or imagined original, where this original itself could be a model.
2. Reduction
   In general, a model will not include all attributes that describe the original but only those that appear relevant to the model's creator or user.
3. Pragmatism
   A model does not relate unambiguously to its original. It is intended to work as a replacement for the original a) for certain subjects (for whom?) b) within a certain time range (when?) c) restricted to certain conceptual or physical actions (what for?).

What is a model?

Mathematics

Mathematical models:

- are simplified descriptions of complex systems expressed in mathematical terms (parameters, variables, equations and inequalities, logical expressions)
- these mathematical terms can be manipulated
- simplifications are based on assumptions
- computers may help to solve these models

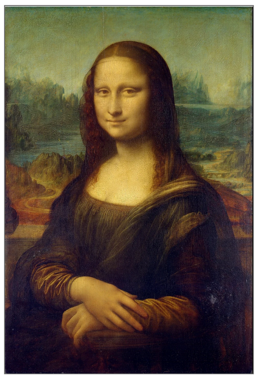
Mathematical modelling is the foundation for deriving models in other natural science and engineering disciplines, such as physics, chemistry, or chemical engineering.

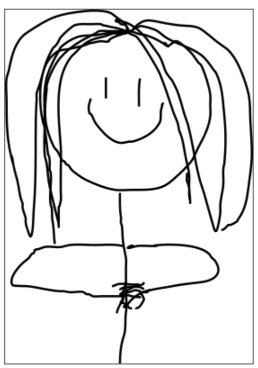
!!!!GRAFIK INPUT OUTPUT MAPPING !!!!

What is a model?

Art

The counterpart to a mathematical or physical model in art is abstraction:





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Why do we need models for sustainable chemical systems?

Examples

- molecular sturcture: geometry optimization
- pure compund properties: boiling temperature, melting temperature, viscosity, toxicity
- predict mixture properties: activity coefficients, solubilities
- reactions: reaction mechanism, reaction rates
- processes: phase equilibria, heat and mass transfer

Why do we need models for sustainable chemical systems?

Useful for:

- gaining an in depth understanding about chemical systems on different scales (from the molecule up to the process system)
- molecular screening and design: finding the optimal molecule for a certain application
- optimizing reactions (space-time yield, conversion, selectivity, atom-efficiency, temperature, pressure, solvent consumption, catalyst loading)
- optimizing processes (cost, energy, environmental impact)
- process design and scale-up
- safety and risk assesment

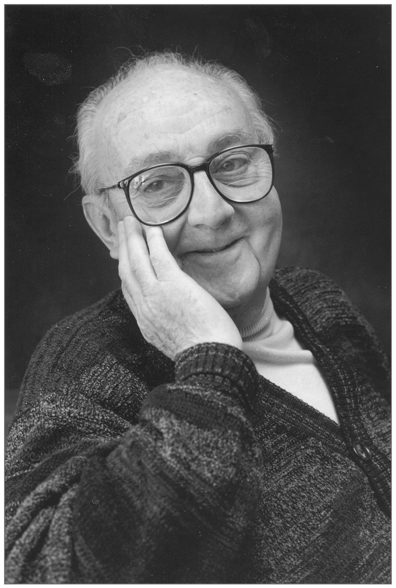
Modelling can make chemical systems more sustainble through optimizing molecules, reactions and processes for minimizing their environmental impact

What makes a good model?

Join at menti.com, use code: 8933 1890

What makes a good model?

"All models are wrong, but some are useful." - George E. P. Box, Statistician



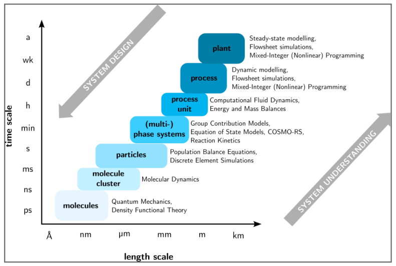
from **wikipedia**, public domain *

What makes a good model?

- captures the essential behavior of the system
- answers the question at hand
- sufficient accuracy
- validated against experiments

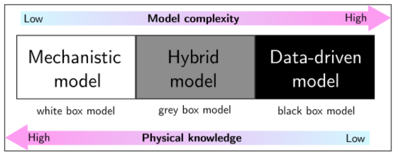
- interpretable
- robust
- computationally practical

Models across the scales



Modelling approches

Mechanistic vs. hybrid vs. data-driven



show on cow example
show on real example

*from **wikipedia** Oeschinensee_Slaunger_2009-07-07.jpg), public domain*

Modelling approaches

Linear vs. nonlinear

Linear structure implies that a problem can be decomposed into simpler parts that can be treated independently or analyzed at a different scale, and therefore that the results will remain valid if the initial is recomposed or rescaled

If all the operators in a mathematical model exhibit linearity, the resulting mathematical model is defined as linear. All other models are considered nonlinear. The definition of linearity and nonlinearity is dependent on context, and linear models may have nonlinear expressions in them. For example, in a statistical linear model, it is assumed that a relationship is linear in the parameters, but it may be nonlinear in the predictor variables. Similarly, a differential equation is said to be linear if it can be written with linear differential operators, but it can still have nonlinear expressions in it. In a mathematical programming model, if the objective functions and constraints are represented entirely by linear equations, then the model is regarded as a linear model. If one or more of the objective functions or constraints are represented with a nonlinear equation, then the model is known as a nonlinear model.

Nonlinearity, even in fairly simple systems, is often associated with phenomena such as chaos and irreversibility. Although there are exceptions, nonlinear systems and models tend to be more difficult to study than linear ones. A common approach to nonlinear problems is linearization, but this can be problematic if one is trying to study aspects such as irreversibility, which are strongly tied to nonlinearity.

Modelling approaches

Dynamic vs. steady-state
A dynamic model accounts for time-dependent changes in the state of the system, while a static (or steady-state) model calculates the system in equilibrium, and thus is time-invariant. Dynamic models are typically represented by differential equations or difference equations. Explicit vs. implicit
If all of the input parameters of the overall model are known, and the output parameters can be calculated by a finite series of computations, the model is said to be explicit. But sometimes it is the output parameters which are known, and the corresponding inputs must be solved for by an iterative procedure, such as Newton's method or Broyden's method. In such a case the model is said to be implicit. For example, a jet engine's physical properties such as turbine and nozzle throat areas can be explicitly calculated given a design thermodynamic cycle (air and fuel flow rates, pressures, and temperatures) at a specific flight condition and power setting, but the engine's operating cycles at other flight conditions and power settings cannot be explicitly calculated from the constant physical properties.

Modelling approaches

Discrete vs. continouus
A discrete model treats objects as discrete, such as the particles in a molecular model or the states in a statistical model; while a continuous model represents the objects in a continuous manner, such as the velocity field of fluid in pipe flows, temperatures and stresses in a solid, and electric field that applies continuously over the entire model due to a point charge. Deterministic vs. probabilistic (stochastic)

Data-driven modelling

AI vs ML vs DL -> Begriffe erklären
supervised vs unsupervised vs reinforcement learning
architectures: types of models (MLP, DNN, GNN, Transformers)